

MOHAMED GEDAN

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 [LinkedIn](#)

SUMMARY

Senior Embedded Software Engineer with 7+ years of experience in low-level embedded systems and microcontroller-based development. Strong background in Embedded C/C++, Embedded Linux, peripheral drivers, real-time systems, and hardware-near debugging. Experienced across the full software development lifecycle, including design, implementation, verification, and system integration. Proven delivery of mission-critical software in globally distributed teams, with hands-on CI/CD automation.

WORK EXPERIENCE

Senior Software Design Engineer

Schneider Electric Automation GmbH | Marktheidenfeld, Germany |

04/2025 – 02/2026

- Led the integration and configuration of EtherCAT for Drive-Controller, as a key project for Schneider Electric Automation.
- Unblocked EtherCAT bring-up by identifying and fixing a critical HW/SW defect in a Xilinx FPGA IP core.
- Authored complete EtherCAT Slave Information (ESI) device profile for the drive SubDevice.
- Built Yocto/PetaLinux for Xilinx Zynq, customizing kernel, **U-Boot**, filesystem, and applications via BitBake.
- Implemented **cybersecurity** Linux service managing **device birth certificate** and **OPC UA FX** based on secure **TPM** storage.
- Contributed to the threat modeling meetings with cybersecurity group.
- Optimized latency of real-time communication using distributed clock synchronization.
- Extended CI/CD pipelines on GitHub Actions to support embedded Linux builds, automated testing, and firmware delivery.

Senior Embedded Software Engineer

SIEMENS s.r.o. | Prague, Czechia |

03/2023 – 01/2025

- Developed hard real-time firmware in C/C++ for industrial communication gateway (CANopen/PROFINET), collaborating with teams across the USA and Germany to deliver a key Siemens Digital Industries project.
- Developed a **secure bootloader(C)** with **signature validation** and **data integrity checks** to enable safe firmware updates.
- Led the development of CI/CD pipelines in Azure DevOps to automate firmware build, test, static code analysis, and release.
- Resolved customers issues of CAN bus installation and defected sensors by design and develop CAN data-logger feature.
- Implemented unit tests (Google Test) and achieved 100% function coverage and 95% line coverage.
- Migrated Linux image builds into Docker based and Bash scripting, enhancing reliability and optimizing image size.

Embedded Software Engineer

Embedded FAB | AlJizah, Egypt |

03/2019 – 02/2023

- Engineered PLCC transceiver algorithms, improving communication stability over power lines.
- Tailored bare-metal drivers for SPI, UART, ADC, and hardware application for STM32F4 microcontroller.
- Built CMake based build systems for ARM toolchain and achieved reliable build system and reduced build time.
- Implemented runtime diagnostic and logging mechanisms via Embedded C to support fault analysis and system stability.

System Engineer

SCEDco (South Cairo Electricity Distribution Company) | Cairo, Egypt |

09/2012 – 02/2019

- Worked in industrial OT environments (SCADA, ADMS, RTUs) as a system and commissioning engineer.
- Integrated and commissioned 65+ SCADA panels and RTUs in operational power networks.
- Applied Scrum/Agile methodology to deliver iterative software increments on time and within scope.

SKILLS & TOOLS

Scripting: Python · Bash (Strong) · Batch

Interfaces: SPI · UART · I2C · CAN · ADC

Firmware Security: Secure Boot · TPM · Certificate Lifecycle Management · OTA Firmware Updates · Threat Modeling

Debugging / Testing: GNU GDB · Google Test (GTest) · Serial Terminal · Wireshark · CAN Bus Analyzer · Green Hills MULTI · J-LINK

Methodologies: Agile · SAFe · Scrum · Kanban · SDLC · Cross-functional collaboration · Technical documentation

CERTIFICATIONS

Certified PROFINET Engineer · 02/2024 · PI training center at ČVUT Prague ([industrial-automation-group](#))

German: A2 (willingness to improve)

EDUCATION

MAM, Faculty of EE. Engineering

Benha University, Faculty of EE. Engineering /

Research paper: DVB-T Spectrum Sensing for Cognitive Radio over Software Defined Radio (SDR).

REFERENCE

Recommendation letter (Empfehlungsschreiben) is available upon request.

LinkedIn recommendation